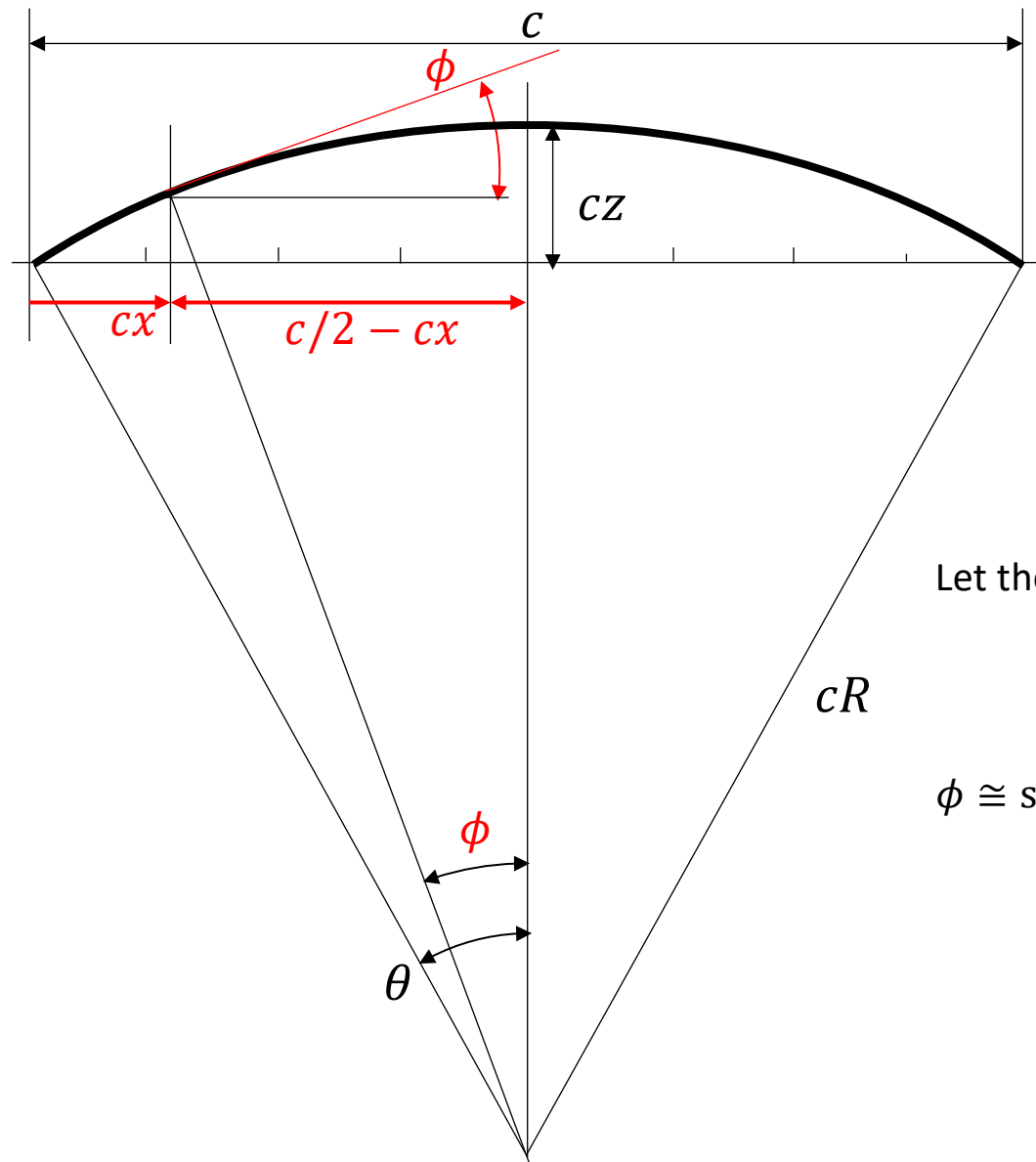


Aeronautical Systems B

Lecture: 1, 2

1. Thin-Wing Theory for Cambered Wings (Arc Wings)
2. Aerodynamic Center

【例】 Arc wing (curvature radius cR , maximum camber ratio z , $\theta \ll 1$, $z \ll 1$)



$$\theta \cong \sin \theta = \frac{c/2}{cR} = \frac{1}{2R} \quad (\text{B2-1})$$

$$\cos \theta \cong 1 - \frac{\theta^2}{2} = 1 - \frac{1}{8R^2} \quad (\text{B2-2})$$

$$cz = cR - cR \cos \theta \quad (\text{B2-3})$$

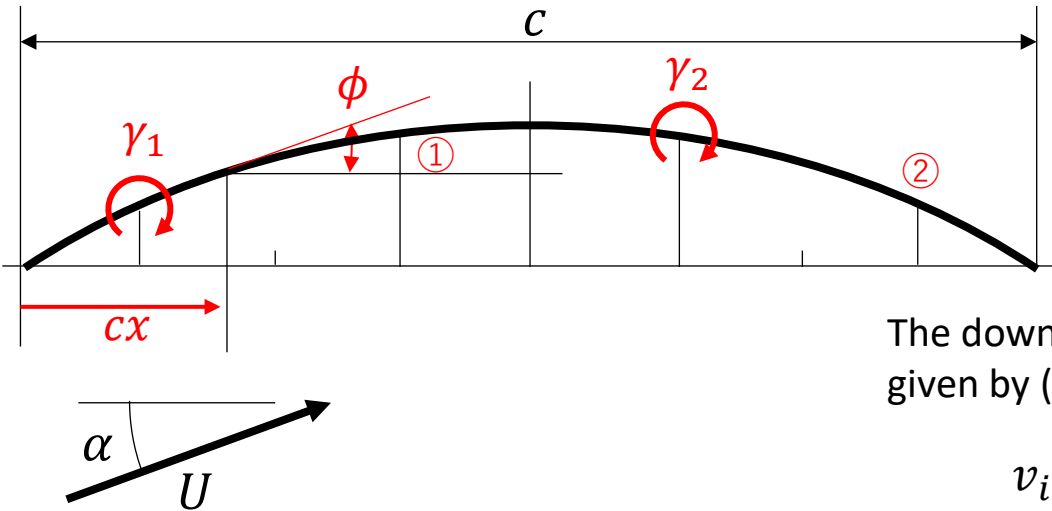
$$z = R - R \cos \theta = \frac{1}{8R} \quad (\text{B2-4})$$

Let the slope of the tangent line at any point cx be ϕ .

$$\phi \cong \sin \phi = \frac{\frac{c}{2} - cx}{cR} = \frac{1 - 2x}{2R} = 4z(1 - 2x) \quad (\text{B2-5})$$

Lift acting on the wing

Divide the wing into two sections, placing vorticities γ_1 and γ_2 at the points $c/8$ and $5c/8$, respectively. Designate the points $3c/8$ and $7c/8$ as control points ① and ②.



$$v_i = \frac{\Gamma}{2\pi r} \quad (\text{B1-12})$$

The downward induction velocity due to γ_1 and γ_2 at ① is given by (B1-12) as

$$v_{i1} = \frac{1}{\pi c} (2\gamma_1 - 2\gamma_1) \quad (\text{B2-9})$$

The downward induction velocity due to γ_1 and γ_2 at ② is

$$v_{i2} = \frac{1}{\pi c} \left(\frac{2}{3}\gamma_1 - 2\gamma_1 \right) \quad (\text{B2-10})$$

Control point conditions ; $v_{n1} = v_{i1}$, $v_{n2} = v_{i2}$

$$\phi_1 = z, v_{n1} = U(\alpha - z) \quad (\text{B2-7})$$

$x = 7/8$ at ②

$$\phi_2 = -3z, v_{n2} = U(\alpha + 3z) \quad (\text{B2-8})$$

$$v_{n1} = U(\alpha - z) = \frac{1}{\pi c} (2\gamma_1 - 2\gamma_2) \quad (\text{B2-11})$$

$$v_{n2} = U(\alpha + 3z) = \frac{1}{\pi c} \left(\frac{2}{3}\gamma_1 + 2\gamma_2 \right) \quad (\text{B2-12})$$

$$U(\alpha - z) = \frac{1}{\pi c} (2\gamma_1 - 2\gamma_2) \quad (\text{B2-11})$$

$$U(\alpha + 3z) = \frac{1}{\pi c} \left(\frac{2}{3}\gamma_1 + 2\gamma_2 \right) \quad (\text{B2-12})$$

Solving this

$$\gamma_1 = \frac{3\pi c U}{4} (\alpha + z) \quad (\text{B2-13})$$

$$\gamma_2 = \frac{\pi c U}{4} (\alpha + 5z) \quad (\text{B2-14})$$

$$L = \rho U \Gamma \quad (\text{B1-13})$$

The lift acting on the point γ_1 from (B1-13) is,

$$L_1 = \rho U \gamma_1 = \frac{3\pi \rho U^2 c}{4} (\alpha + z) \quad (\text{B2-15})$$

The lift acting on point γ_2 is

$$L_2 = \rho U \gamma_2 = \frac{\pi \rho U^2 c}{4} (\alpha + 5z) \quad (\text{B2-16})$$

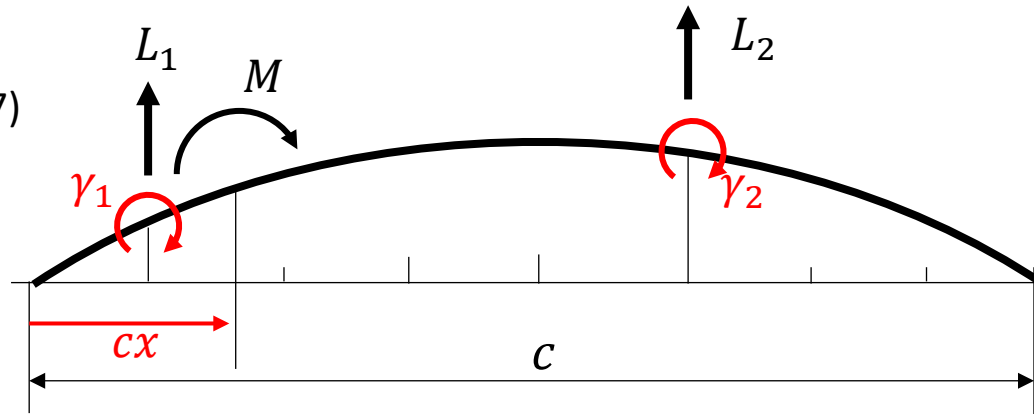
Lift acting on the wing

The lift acting on the entire wing

$$L = \rho U(\gamma_1 + \gamma_2) = \pi \rho U^2 c(\alpha + 2z) \quad (\text{B2-17})$$

Make this dimensionless

$$C_l = \frac{L}{\frac{1}{2} \rho U^2 c} = 2\pi(\alpha + 2z) \quad (\text{B2-18})$$



Furthermore, the nose-up moment acting on a point at a distance cx from the leading edge is

(B1-13)

$$M = (cx - c/8)L_1 + (cx - 5c/8)L_2 \quad (\text{B2-19})$$

Make this dimensionless

$$C_m = \frac{M}{\frac{1}{2} \rho U^2 c^2} = 2(x - 1/8) \frac{3}{4} \pi(\alpha + z) + 2(x - 5/8) \frac{\pi}{4}(\alpha + 5z) = \frac{\pi}{2} \left[(4x - 1)\alpha + \left(8x - \frac{7}{2} \right) z \right]$$

(B2-20)

Lift coefficient $C_l = 2\pi(\alpha + 2z)$ (B2-18)

The lift gradient is $(\partial C_l)/\partial \alpha = 2\pi$ and remains unchanged even if the airfoil has camber.

Moment coefficient $C_m = \frac{\pi}{2} \left[(4x - 1)\alpha + \left(8x - \frac{7}{2} \right) z \right]$ (B2-20)

The moment coefficient around the point $x = 1/4$ (the 1/4-chord point) is

$$C_{m_{1/4}} = -\frac{3\pi z}{4} \quad (\text{B2-21})$$

This remains constant regardless of the angle of attack. The point where the moment does not change with the angle of attack is called the **aerodynamic center**. For a flat plate ($z=0$), $C_{m_{1/4}} = 0$.